Welcome to the case study!

PYTHON TOOLBOX

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World bank data

- Data on world economies for over half a century
- Indicators
  - Population
  - Electricity consumption
  - CO2 emissions
  - Literacy rates
  - Unemployment
  - Mortality rates

Using zip()

```
avengers = ['hawkeye', 'iron man', 'thor', 'quicksilver']
names = ['barton', 'stark', 'odinson', 'maximoff']
z = zip(avengers, names)
print(type(z))
```

```
<class 'zip'>
```

```
print(list(z))
```

```
[('hawkeye', 'barton'), ('iron man', 'stark'),
('thor', 'odinson'), ('quicksilver', 'maximoff')]
```

Defining a function

- raise.py

```
def raise_both(value1, value2):
    """Raise value1 to the power of value2
    and vice versa."""
    new_value1 = value1 ** value2
    new_value2 = value2 ** value1
    new_tuple = (new_value1, new_value2)
    return new_tuple
```

Re-cap: list comprehensions

Basic

```
[output expression for iterator variable in iterable]
```

Advanced

```
[output expression +
conditional on output for iterator variable in iterable +
conditional on iterable]
```

Let's practice!

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Using Python generators for streaming data

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Generators for the large data limit

- Use a generator to load a file line by line
- Works on streaming data!
- Read and process the file until all lines are exhausted

Build a generator function

- sequence.py

```
def num_sequence(n):
    """Generate values from 0 to n."""
    i = 0
    while i < n:
        yield i
        i += 1
```

Let's practice!

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Using pandas' read_csv iterator for streaming data

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Reading files in chunks

- Up next:
  - `read_csv()` function and `chunksize` argument
  - Look at specific indicators in specific countries
  - Write a function to generalize tasks

Let's practice!

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Final thoughts

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You have applied your skills in:

- User-defined functions
- Iterators
- List comprehensions
- Generators

Congratulations!

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